

CSR PROJECT BY COMPANIES

Project Overview

This project analyzes Corporate Social Responsibility (CSR) projects undertaken by companies across India. The source dataset contains project-level information such as year, state, district, development sector, company details, project outlay and amount spent. Excel/CSV is used as the source data for inspection and preparation, while Python and Power BI are used to build an interactive analytical dashboard.

The project is designed to convert raw CSR records into meaningful business and social-impact insights. The analysis focuses on project volume, geographical distribution, development sectors, company participation, planned outlay, actual spending and spending efficiency.

Project Objectives

- Understand the distribution and characteristics of CSR projects undertaken by companies.
- Measure CSR project activity across years, states, districts, companies and development sectors.
- Compare project outlay with actual CSR spending to understand utilization.
- Identify major contributing companies, states and CSR development sectors.
- Create meaningful KPIs and interactive Power BI visualizations for decision-making.
- Generate clear business/social-impact insights from the cleaned dataset.

Project Scope

Scope Area	Description
1. Data Loading and Initial Overview	Load the Excel/CSV data, inspect structure, identify columns, data types, row count, column count, missing values and duplicates.
2. Data Pre-processing	Clean missing values, verify data types, remove duplicate records where required, standardize fields and create analysis-ready fields.
3. Exploratory Data Analysis (EDA)	Study distributions, categories, trends, company participation, geography, sectors and spending patterns.
4. Visualizations	Create charts and KPI cards using Power BI to communicate patterns and comparisons interactively.

Project Domains

Domain	Application
Primary Domain	Corporate Social Responsibility (CSR) / Social Impact
Analytics Domain	Data Analytics and Business Intelligence
Business Domain	Corporate Finance, CSR Spending and Company Performance
Geographical Domain	State, district and India-level CSR project distribution
Technology Domain	Microsoft Excel, Python/Pandas and Microsoft Power BI

Dataset Information

The supplied dataset is a project-level CSR dataset containing records of CSR activities by companies. Each row represents a CSR project record and the columns describe its time, location, project sector, company characteristics and financial amounts.

Dataset Metric	Value
Rows / Records	36,664
Columns / Features	17
Unique Companies	693
Unique States / Regions	37
Development Sectors	29
Company Categories	3
Year Range	2014–2020
Duplicate Rows	0
Missing Values	3 cells

Dataset Features

Column	Meaning	Category
id	Unique project record identifier	Identifier
year	CSR project year/date	Time
state_name / state_code	State or region information	Geography
district_name / district_code	District information	Geography
csr_project_name	Name/description of CSR project	Project
development_sector	CSR development area	Category
company_name	Company undertaking the project	Company
cin	Corporate Identification Number	Company ID
company_class	Public/private/PLC classification	Company
company_type	Government or other company type	Company
roc	Registrar of Companies	Company/Location
company_category	Legal/company category	Company
listing_status	Listed or unlisted status	Company
project_outlay_amount_in_lakhs	Planned/project outlay in lakh rupees	Financial
amount_spent_in_lakhs	Actual amount spent in lakh rupees	Financial

Data Loading and Initial Overview

The dataset was loaded from the provided CSV/Excel-based source and reviewed before visualization. The year field was interpreted as a date field and financial columns were treated as numeric measures.

- Dataset shape: 36,664 rows × 17 columns.
- Column names and data types were reviewed before analysis.
- Duplicate records found: 0.
- Missing cells found: 3; the missing values are concentrated in the CSR project name field.
- Financial fields were checked as numeric measures for aggregation in Power BI.
- Year values cover 2014 through 2020.

Data pre-processing

- Checked missing values and identified the small number of missing CSR project-name entries.
- Checked duplicate rows; the supplied dataset contains no exact duplicate rows.
- Converted/validated the year field as a date and extracted year information for trend analysis.
- Validated project outlay and amount-spent fields as numeric values.
- Retained the original business dimensions such as state, district, company and development sector for Power BI slicing.
- Created/used analytical measures in Power BI such as Total CSR Projects, Total Companies, Total CSR Outlay, Total Amount Spent, Average Amount Spent and Spending Efficiency %.
- A log-transformed project-outlay field is present in the Power BI model for handling highly skewed financial values.

Exploratory Data Analysis (EDA)

EDA Measure	Result
Total project records	36,664
Companies represented	693
States/regions represented	37
Development sectors	29
Total project outlay	₹45,459,538,227.04 lakh
Total amount spent	₹5,703,598.21 lakh
Overall spent/outlay ratio	0.0125%
Average amount spent per record	₹155.56 lakh

Major Geographical Pattern

Maharashtra has the highest number of project records in the supplied data, followed by the All-India category, Gujarat, Tamil Nadu and Karnataka. This indicates that CSR activity is strongly represented in major industrial and corporate locations.

Rank	State / Region	Project Records
1	Maharashtra	7,037
2	All India	6,749
3	Gujarat	3,491
4	Tamil Nadu	2,552
5	Karnataka	1,933

Year-wise Project Activity

Year	Project Records
2014	3,268
2015	4,740
2016	5,136
2017	5,285
2018	6,022
2019	6,234
2020	5,979

Project-record volume rises from 3,268 in 2014 to a peak of 6,234 in 2019, followed by a small reduction in 2020. The pattern can be explored further in Power BI using year filters and trend visuals.

Power BI Dashboard

The supplied PBIX file was inspected and contains three report pages. The dashboard uses KPI cards, pie/donut charts, line and column charts, area charts, a treemap, gauge, stacked visuals and slicers.

Power BI Page	Purpose
OVER VIEW	High-level KPI summary and overall CSR performance.
PROJECT ANALYSIS	Project, state, sector, company-category and time-based analysis.
COMPANY EXPENSE	Company-level spending, outlay and spending-efficiency analysis.

Power BI Visual Components Identified

- KPI cards — 6 visual(s)
- Pie charts — 2 visual(s)
- Line chart — 1 visual(s)
- Bar chart — 1 visual(s)
- image — 1 visual(s)
- textbox — 1 visual(s)
- Slicer/filter — 1 visual(s)
- Area chart — 1 visual(s)
- Donut chart — 1 visual(s)
- Line + clustered column combination chart — 1 visual(s)
- 100% stacked area chart — 1 visual(s)
- Treemap — 1 visual(s)
- Stacked area chart — 1 visual(s)
- Gauge — 1 visual(s)
- 100% stacked column chart — 1 visual(s)
- Clustered column chart — 1 visual(s)

Key Power BI Measures

Measure	Purpose
Total CSR Projects	Count of CSR project records.
Total Companies	Distinct company count.
Total CSR Outlay	Sum of project outlay amount.
Total Amount Spent	Sum of actual amount spent.
Average Amount Spent	Average actual spending per record.
Spending Efficiency %	Actual amount spent compared with project outlay.

Visualization Interpretation

KPI Cards

KPI cards provide a quick summary of total CSR projects, participating companies, total planned outlay, total amount spent, average spending and spending efficiency. They are useful for management-level monitoring.

Year-wise Line Chart

The year trend shows how CSR project spending/activity changes over time. It helps identify periods of increasing or decreasing project activity and supports year-on-year comparison.

State-wise Bar Chart

The state-wise comparison highlights where CSR projects are concentrated geographically. Higher project counts can indicate stronger corporate presence or greater CSR activity in those regions.

Development-sector Donut/Pie Chart

The distribution sector shows which CSR themes receive the greatest project attention. Education and health-related activities form major portions of the project records.

Company-category Pie Chart

The company-category visual helps compare the composition of companies participating in CSR projects and supports analysis of the legal/company structure represented in the data.

Outlay vs Amount Spent Visual

Comparing planned outlay with actual spending helps evaluate financial utilization and identify categories or sectors where the gap is larger.

Company Spending Visual Efficiency

The company-level efficiency analysis helps identify companies with relatively higher or lower spending against their planned outlay.

Treemap / Geographic Analysis

The treemap supports exploration of project distribution by district and project name, allowing users to identify concentrated CSR activity at a more detailed geographical level.

Tools and Technologies

Tool	Use in Project
Microsoft Excel / CSV	Dataset storage, inspection and initial data review
Python / Pandas	Data loading, validation, cleaning and EDA
Microsoft Power BI	Dashboard creation, measures, filters and interactive visualizations
Power BI Data Model / DAX	KPI measures and analytical calculations

Conclusion

The CSR Project by Companies analysis combines structured data preparation with exploratory analysis and Power BI visualization to understand corporate social responsibility activity in India. The project identifies where CSR projects are concentrated, which development sectors receive significant attention, how project activity changes over time, and how planned outlay compares with actual spending. The resulting Power BI dashboard provides an interactive and practical way to explore CSR project performance.